VIDYAVAHINI GROUP OF INSTITUTIONS

VIDYAVAHINI FIRST GRADE & P.G. COLLEGE

Affiliated to Tumkur University

Project Report

On

MINI PROJECT

"Submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications."

Submitted By

Ananya M P U11VU24S0036

Bhoomika U11VU24S0349

Manasa K S U11VU24S0070

Shubha C S U11VU24S0304

Under the Guidance of

Navyashree S

Department of Computer Science

DEPARTMENT OF COMPUTER APPLICATIONS VIDYAVAHINI FIRST GRADE & P.G. COLLEGE TUMAKURU

1. Title of the Project

AI Virtual Hand Cursor Using Hand Gesture Recognition

2. Abstract / Introduction to Project

The **AI Virtual Hand Cursor** is a computer vision-based system designed to control basic computer mouse operations using hand gestures. The system uses a webcam to capture the user's hand movements and processes the video using Python, OpenCV, and MediaPipe. Hand landmarks are detected and tracked in real time to identify finger positions and specific gestures. These gestures are then converted into mouse actions such as cursor movement, left-click, right-click, and scrolling using PyAutoGUI. The project provides a touch-free and interactive method of human-computer interaction, reducing the dependency on a traditional physical mouse.

3. Existing System

Problem Statement: Traditional computer systems primarily depend on a physical mouse for cursor movement and interaction. This requires continuous physical contact and may not always be convenient in situations where touch-free interaction is preferred.

Disadvantages:

- Requires a physical mouse for interaction.
- Requires continuous hand contact with the device.
- Physical mouse devices can experience hardware failures.
- Provides limited natural gesture-based interaction.
- May not be convenient in certain touch-free environments.

4. Proposed System

Solution: The proposed system uses Artificial Intelligence and Computer Vision to create a virtual mouse controlled through hand gestures. A webcam captures the user's hand, and MediaPipe detects hand landmarks. The system analyzes finger positions and recognizes predefined gestures. The recognized gestures are mapped to mouse operations using PyAutoGUI.

Advantages:

• Provides touch-free computer interaction.

• Reduces dependency on a physical mouse.

• Uses a standard webcam for input.

• Provides real-time hand tracking.

• Easy to use and demonstrate.

• Can be extended with additional gestures.

• Uses widely available software technologies.

5.Modules

No.	Module	Description
1	Video Capture	Captures live video from the webcam and provides frames for processing.
2	Hand Detection	Detects the user's hand using the MediaPipe hand-tracking framework.
3	Hand Landmark Tracking	Identifies important points on the hand, including fingertips and finger joints.
4	Gesture Recognition	Analyzes finger positions to identify predefined hand gestures.
5	Cursor Control	Maps index-finger movement to the computer screen and controls the virtual cursor.
6	Mouse Action Control	Converts recognized gestures into operations such as left-click, right-click, and scrolling.
7	Visual Feedback	Displays the camera feed and hand landmarks so users can observe detected gestures.

6. Hardware and Software Requirements

Sl. No.	Hardware	Requirement
1	Computer/Laptop	Minimum 4 GB RAM
2	Processor	Intel Core i3 or equivalent
3	Webcam	Built-in or USB webcam
4	Storage	Minimum 500 MB free space
5	Display	Standard monitor/laptop display

7.Software Requirements

Sl. No.	Software	Purpose
1	Python	Main programming language
2	OpenCV	Image and video processing
3	MediaPipe	Hand detection and landmark tracking
4	PyAutoGUI	Mouse and cursor control
5	NumPy	Numerical and coordinate calculations
6	Windows/Linux	Operating system

Team Members Signatures: **Project Guide Signature:**

1.

2.

3.

4.

Signature of the HOD